

Week 5: Algorithmic Refinement and Data Cleaning

Hybrid QML and Quantum String Algorithms for Document Extraction

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Deliverable as specified in the project brief: Cleaned data output ready for industrial application.

1 Status: Not Delivered

This deliverable was not completed. This report states what was specified, why it was cut, what would have been required, and what was done instead with the time.

A documented account of an attempted and abandoned objective is a legitimate part of a research report; silently omitting the week would not be.

2 What Was Specified

The project brief set out three items for this week:

- Implement quantum subroutines for string alignment and cleaning of OCR noise.
- Replace LLM-based parsing with targeted quantum logic for specific data schemas.
- Deliver cleaned data output ready for industrial application.

3 Why It Was Cut

3.1 The Precondition Was Not Met

Quantum string alignment operates on extracted text. Its purpose is to correct residual OCR errors—aligning a noisy read against a schema or a reference vocabulary and repairing mismatches.

That presupposes an extraction stage whose output is mostly correct, with sparse and localised errors. The measured reality at the point this week began:

Table 1: Extraction quality available as input to a cleaning stage.

Tier	CER	Identifier recovered exactly
clean_digital	6.6%	100%
clean_scan	8.5%	58%
noisy_scan	79.5%	0%
clear_handwriting	51.8%	0%
degraded_handwriting	85.7%	0%

On the two clean tiers, cleaning has little to work with: 100% of clean digital identifiers are already exact, so a cleaning stage could improve nothing there. On the three degraded tiers, error rates of 52–86% mean the input is not noisy text but largely wrong text. Alignment against a schema does not repair a string where most characters are incorrect; it produces a confident and wrong result.

3.2 The Resource Cost Was Prohibitive

Quantum sequence alignment via a Grover-based approach would extend the comparator oracle of Week 4. That oracle’s measured cost is the obstacle:

- 3,272 CX gates for a 16-character field with a 2-character pattern.
- Measured scaling exponent of 1.39 in text length—superlinear, because each candidate position requires its own controlled data load.
- A full 38-symbol alphabet needs 6 qubits per character, placing even short alignment windows beyond feasible simulation.

Alignment requires comparing against multiple reference candidates rather than a single pattern, multiplying an already superlinear cost. Week 4 had already established that the $\sqrt{N}$ advantage does not survive data loading for single-pattern search; there was no basis to expect it to survive for alignment.

3.3 Time

The project ran on a compressed schedule. Section ?? records the circumstances. Given a finite budget, the choice was between attempting this deliverable and completing the benchmarking that Week 6 specified.

Attempting this week would have produced a component that did not work, resting on an extraction stage that could not support it, at a resource cost already measured as prohibitive. A broken feature discovered at submission is worse than a documented decision not to build it.

4 What Would Have Been Required

For completeness, the path to delivering this week properly:

1. **Fix the extraction stage first.** The measured bottleneck is the classical segmentation front end, not the quantum stages. Clean-tier CER of 6.6–8.5% against 93–95% segmentation recovery means the residual error is missed and merged glyphs. A learned detector replacing projection profiles is the correct first step, and would raise extraction quality to the point where cleaning has something to act on.
2. **Adopt a coarser encoding for search.** The one-qubit-per-candidate-position encoding is what makes the oracle expensive. Stem-level or block-level encodings, as used in related QUBO formulations, reduce both qubit count and loading cost.
3. **Define the schema constraint explicitly.** “Targeted quantum logic for specific data schemas” requires a specified schema. The synthetic corpus uses a fixed identifier format, which would be a reasonable starting constraint, but the brief did not specify a target schema and none was agreed.
4. **Establish a classical baseline first.** Fuzzy matching against a schema is a solved classical problem. Without measuring that baseline, there would be no way to tell whether a quantum implementation contributed anything—which is exactly the failure mode Weeks 2 and 4 were designed to avoid.

5 What Was Done Instead

The time was spent on work not specified in the brief but which strengthened the project’s core claims:

- **Statistical significance testing.** The Week 2 comparison rested on a single train–test split. Repeating over 30 paired splits established that the quantum-versus-classical result is a null result ($p \approx 0.2\text{--}0.3$) while the raw-pixel lead is real ($p < 10^{-10}$). Without this the two claims were indistinguishable.
- **An external baseline.** Tesseract was run on identical images, providing the reference point that the internal comparisons lacked.
- **Noise and hardware-feasibility analysis.** Gate error, readout error and shot noise were modelled, establishing that the feature extractor is plausible on near-term hardware while the search stage is not.
- **Reproducibility infrastructure.** Committed dataset and model, browser-parity verification, and continuous integration that fails when the report and measured results disagree.

6 Scope

The project was scoped for three participants, with separate owners for the vision and search tracks. Mentor contact ended in early July, and the two track assignments were not taken up. Both tracks were consolidated into a single-person scope.

This is recorded as context for the schedule, not as justification for the decision above. The precondition and resource arguments of Section 3 stand independently: this deliverable would have been the correct one to cut even with a full team.

7 Deliverable Status

Not delivered. Cut deliberately, with reasons recorded here and in Section 8 of the final report.